Chapter 9

Differentiation

Real-Life Connection

Differentiation helps us understand change.
It is used in:

- Speed and acceleration
- Population growth
- Profit and cost analysis
- Physics and Engineering
- Artificial Intelligence
- Medical science

Whenever quantities change continuously, differentiation becomes useful.

9.1 Overview

Differentiation is one of the most important concepts in mathematics.

It measures the rate at which one quantity changes with respect to another quantity.

If:

$$y = f(x)$$

then the derivative tells us how fast y changes when x changes.

Differentiation is widely used in:

- Calculus
- Physics
- Engineering
- Economics

- Machine Learning
- Optimization problems

9.2 Why Should We Study Differentiation?

Importance of Differentiation

1. To calculate instantaneous speed
2. To find maximum profit and minimum cost
3. To study motion and acceleration
4. To analyze graphs
5. To solve optimization problems
6. To understand scientific models

9.3 Definition of Derivative

Suppose:

$$y = f(x)$$

Then the derivative is written as:

$$\frac{dy}{dx} \quad \text{or} \quad f'(x)$$

Derivative represents:

- Rate of change
- Slope of tangent
- Instantaneous variation

9.4 Differentiation Using First Principle

9.4.1 Formula

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

This is called the first principle definition of derivative.

9.4.2 Steps

1. Replace x by $x + h$
2. Find $f(x + h)$
3. Substitute into formula
4. Simplify
5. Apply limit $h \to 0$

9.5 Solved Examples

Example 1

Find derivative of:

$$f(x) = x^2$$

using first principle.
Solution

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$f(x+h) = (x+h)^2$$

$$= x^2 + 2xh + h^2$$

Substitute:

$$= \lim_{h \to 0} \frac{x^2 + 2xh + h^2 - x^2}{h}$$

$$= \lim_{h \to 0} \frac{2xh + h^2}{h}$$

$$= \lim_{h \to 0} (2x + h)$$

$$= 2x$$

$$\boxed{f'(x) = 2x}$$

9.6 Basic Differentiation Formulas

Function	Derivative
c	0
x	1
x^n	nx^{n-1}
$\sqrt{x}$	$\frac{1}{2\sqrt{x}}$
$\frac{1}{x}$	$-\frac{1}{x^2}$
e^x	e^x
$\ln x$	$\frac{1}{x}$
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$\sec^2 x$

9.7 Algebra of Differentiation

9.7.1 Sum Rule

If:

$$y = u(x) + v(x)$$

then:

$$\frac{dy}{dx} = u'(x) + v'(x)$$

Example

Differentiate:

$$y = x^2 + 3x$$

$$\frac{dy}{dx} = 2x + 3$$

9.7.2 Difference Rule

If:

$$y = u(x) - v(x)$$

then:

$$\frac{dy}{dx} = u'(x) - v'(x)$$

Example

Differentiate:

$$y = x^3 - 5x$$

$$\frac{dy}{dx} = 3x^2 - 5$$

9.7.3 Product Rule

If:

$$y = u(x)v(x)$$

then:

$$\frac{dy}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}$$

Example

Differentiate:

$$y = x^2(x+1)$$

Let:

$$u = x^2, \quad v = x + 1$$

Then:

$$u' = 2x, \quad v' = 1$$

$$\frac{dy}{dx} = x^2(1) + (x+1)(2x)$$

$$= 3x^2 + 2x$$

9.7.4 Quotient Rule

If:

$$y = \frac{u(x)}{v(x)}$$

then:

$$\frac{dy}{dx} = \frac{vu' - uv'}{v^2}$$

Example

Differentiate:

$$y = \frac{x^2+1}{x}$$

Let:

$$u = x^2 + 1, \quad v = x$$

$$u' = 2x, \quad v' = 1$$

$$\frac{dy}{dx} = \frac{x(2x) - (x^2+1)}{x^2}$$

$$= \frac{x^2-1}{x^2}$$

9.8 Chain Rule

If:

$$y = f(g(x))$$

then:

$$\frac{dy}{dx} = f'(g(x)) \cdot g'(x)$$

Example

Differentiate:

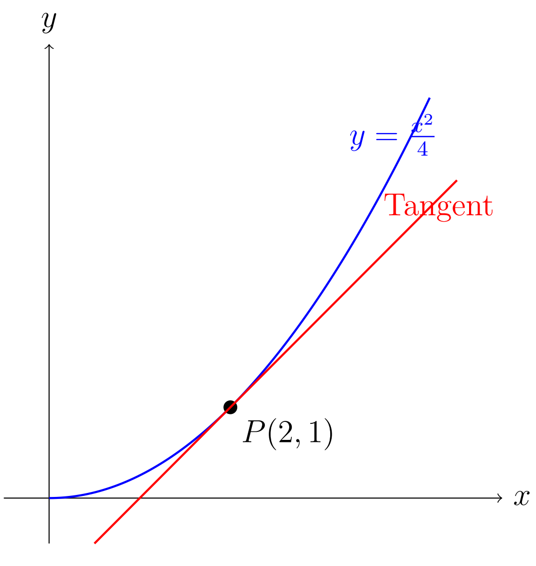
$$y = (x^2 + 1)^5$$

$$\frac{dy}{dx} = 5(x^2 + 1)^4(2x)$$

$$= 10x(x^2 + 1)^4$$

9.9 Geometric Interpretation of Derivative

The derivative of a function at a point represents the slope of the tangent to the curve at that point.



Interpretation:

- Positive derivative $\Rightarrow$ Increasing function
- Negative derivative $\Rightarrow$ Decreasing function
- Zero derivative $\Rightarrow$ Stationary point

9.10 Graphical Behaviour of a Function and Its Derivative

Consider the function:

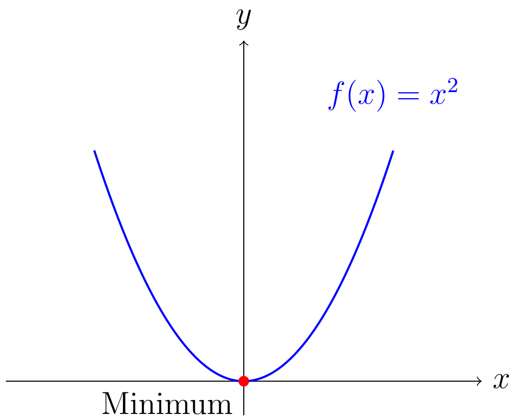
$$f(x) = x^2$$

Its derivative is:

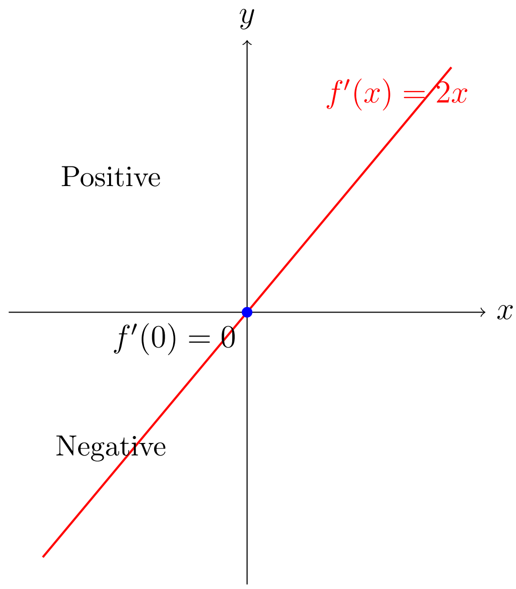
$$f'(x) = 2x$$

Observation:

- The graph of $f(x) = x^2$ is a parabola.
- The derivative $f'(x) = 2x$ gives the slope of the curve at each point.
- When $x < 0$, derivative is negative $\Rightarrow$ function decreases.
- When $x > 0$, derivative is positive $\Rightarrow$ function increases.
- At $x = 0$, derivative is zero $\Rightarrow$ minimum point occurs.



Original Function



Derivative Function

Graph Interpretation

- The original function $f(x) = x^2$ decreases first and then increases.
- The derivative graph explains this behavior.
- Negative derivative $\Rightarrow$ decreasing function.
- Positive derivative $\Rightarrow$ increasing function.
- Zero derivative indicates a turning point.

9.11 Applications of Differentiation

Applications

1. Velocity and acceleration
2. Business optimization
3. Marginal cost and revenue
4. Curve sketching
5. Engineering analysis
6. Machine learning algorithms
7. Population growth studies

9.12 Higher Order Derivatives

If:

$$\frac{dy}{dx} = f'(x)$$

then the derivative of $f'(x)$ is called second derivative:

$$\frac{d^2y}{dx^2}$$

Example

If:

$$y = x^3$$

then:

$$\frac{dy}{dx} = 3x^2$$

Again differentiating:

$$\frac{d^2y}{dx^2} = 6x$$

9.13 Common Mistakes

Mistake Alert

- Forgetting chain rule
- Sign mistakes
- Incorrect simplification
- Ignoring denominator square in quotient rule
- Confusing function and derivative

9.14 Differentiation by First Principle

The first principle definition of derivative is:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

9.14.1 Derivative of x^n

Let:

$$f(x) = x^n$$

Using first principle,

$$f'(x) = \lim_{h \to 0} \frac{(x+h)^n - x^n}{h}$$

Using Binomial Theorem:

$$(x+h)^n = x^n + nx^{n-1}h + \cdots$$

Substituting,

$$= \lim_{h \to 0} \frac{x^n + nx^{n-1}h + \cdots - x^n}{h}$$

$$= \lim_{h \to 0} \left(nx^{n-1} + \cdots\right)$$

As $h \to 0$,

$$\boxed{\frac{d}{dx}(x^n) = nx^{n-1}}$$

9.14.2 Derivative of $\sin x$

Let:

$$f(x) = \sin x$$

Using first principle,

$$f'(x) = \lim_{h \to 0} \frac{\sin(x+h) - \sin x}{h}$$

Using identity:

$$\sin(x+h) = \sin x \cos h + \cos x \sin h$$

Substitute:

$$= \lim_{h \to 0} \frac{\sin x \cos h + \cos x \sin h - \sin x}{h}$$

$$= \lim_{h \to 0} \left[\sin x \cdot \frac{\cos h - 1}{h} + \cos x \cdot \frac{\sin h}{h} \right]$$

Using limits:

$$\lim_{h \to 0} \frac{\sin h}{h} = 1$$

$$\lim_{h \to 0} \frac{\cos h - 1}{h} = 0$$

Therefore,

$$\boxed{\frac{d}{dx}(\sin x) = \cos x}$$

9.14.3 Derivative of $\cos x$

Let:

$$f(x) = \cos x$$

Using first principle,

$$f'(x) = \lim_{h \to 0} \frac{\cos(x+h) - \cos x}{h}$$

Using identity:

$$\cos(x+h) = \cos x \cos h - \sin x \sin h$$

Substitute:

$$= \lim_{h \to 0} \frac{\cos x \cos h - \sin x \sin h - \cos x}{h}$$

$$= \lim_{h \to 0} \left[\cos x \cdot \frac{\cos h - 1}{h} - \sin x \cdot \frac{\sin h}{h} \right]$$

Using standard limits,

$$= 0 - \sin x$$

Hence,

$$\boxed{\frac{d}{dx}(\cos x) = -\sin x}$$

9.14.4 Derivative of $\tan x$

Let:

$$f(x) = \tan x$$

Since:

$$\tan x = \frac{\sin x}{\cos x}$$

Using quotient rule,

$$\frac{d}{dx}(\tan x) = \frac{\cos x(\cos x) - \sin x(-\sin x)}{\cos^2 x}$$

$$= \frac{\cos^2 x + \sin^2 x}{\cos^2 x}$$

Using identity:

$$\sin^2 x + \cos^2 x = 1$$

Therefore,

$$\boxed{\frac{d}{dx}(\tan x) = \sec^2 x}$$

9.14.5 Derivative of e^x

Let:

$$f(x) = e^x$$

Using first principle,

$$f'(x) = \lim_{h \to 0} \frac{e^{x+h} - e^x}{h}$$

$$= \lim_{h \to 0} \frac{e^x e^h - e^x}{h}$$

$$= e^x \lim_{h \to 0} \frac{e^h - 1}{h}$$

Using standard limit:

$$\lim_{h \to 0} \frac{e^h - 1}{h} = 1$$

Therefore,

$$\boxed{\frac{d}{dx}(e^x) = e^x}$$

9.14.6 Derivative of $\log x$

Let:

$$f(x) = \log x$$

Using first principle,

$$f'(x) = \lim_{h \to 0} \frac{\log(x+h) - \log x}{h}$$

Using logarithm property:

$$= \lim_{h \to 0} \frac{\log\left(\frac{x+h}{x}\right)}{h}$$

$$= \lim_{h \to 0} \frac{\log\left(1 + \frac{h}{x}\right)}{h}$$

Let:

$$u = \frac{h}{x}$$

Then:

$$h = ux$$

Substitute:

$$= \lim_{u \to 0} \frac{\log(1+u)}{ux}$$

$$= \frac{1}{x} \lim_{u \to 0} \frac{\log(1+u)}{u}$$

Using standard limit:

$$\lim_{u \to 0} \frac{\log(1+u)}{u} = 1$$

Therefore,

$$\boxed{\frac{d}{dx}(\log x) = \frac{1}{x}}$$

9.15 Solved Examples on Product Rule

Recall:

$$\frac{d}{dx}[uv] = u\frac{dv}{dx} + v\frac{du}{dx}$$

Example 1

Differentiate:

$$y = x^2(x + 3)$$

Solution
Let:

$$u = x^2, \quad v = x + 3$$

Then:

$$u' = 2x, \quad v' = 1$$

Using product rule:

$$\frac{dy}{dx} = x^2(1) + (x + 3)(2x)$$

$$= x^2 + 2x^2 + 6x$$

$$= 3x^2 + 6x$$

$$\boxed{\frac{dy}{dx} = 3x^2 + 6x}$$

Example 2

Differentiate:

$$y = (2x + 1)(x^3)$$

Solution
Let:

$$u = 2x + 1, \quad v = x^3$$

Then:

$$u' = 2, \quad v' = 3x^2$$

Applying product rule:

$$\frac{dy}{dx} = (2x + 1)(3x^2) + x^3(2)$$

$$= 6x^3 + 3x^2 + 2x^3$$

$$= 8x^3 + 3x^2$$

$$\boxed{\frac{dy}{dx} = 8x^3 + 3x^2}$$

Example 3

Differentiate:

$$y = (x^2 + 1)(x^2 - 1)$$

Solution
Let:

$$u = x^2 + 1, \quad v = x^2 - 1$$

Then:

$$u' = 2x, \quad v' = 2x$$

Using product rule:

$$\frac{dy}{dx} = (x^2 + 1)(2x) + (x^2 - 1)(2x)$$

$$= 2x(x^2 + 1 + x^2 - 1)$$

$$= 2x(2x^2)$$

$$= 4x^3$$

$$\boxed{\frac{dy}{dx} = 4x^3}$$

9.16 Solved Examples on Quotient Rule

Recall:

$$\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{vu' - uv'}{v^2}$$

Example 1

Differentiate:

$$y = \frac{x^2 + 1}{x}$$

Solution
Let:

$$u = x^2 + 1, \quad v = x$$

Then:

$$u' = 2x, \quad v' = 1$$

Applying quotient rule:

$$\frac{dy}{dx} = \frac{x(2x) - (x^2 + 1)(1)}{x^2}$$

$$= \frac{2x^2 - x^2 - 1}{x^2}$$

$$= \frac{x^2 - 1}{x^2}$$

$$\boxed{\frac{dy}{dx} = \frac{x^2 - 1}{x^2}}$$

Example 2

Differentiate:

$$y = \frac{x^3 + 2}{x^2}$$

Solution
Let:

$$u = x^3 + 2, \quad v = x^2$$

Then:

$$u' = 3x^2, \quad v' = 2x$$

Using quotient rule:

$$\frac{dy}{dx} = \frac{x^2(3x^2) - (x^3 + 2)(2x)}{x^4}$$

$$= \frac{3x^4 - 2x^4 - 4x}{x^4}$$

$$= \frac{x^4 - 4x}{x^4}$$

$$= 1 - \frac{4}{x^3}$$

$$\boxed{\frac{dy}{dx} = 1 - \frac{4}{x^3}}$$

Example 3

Differentiate:

$$y = \frac{x^2 - 1}{x + 1}$$

Solution
Let:

$$u = x^2 - 1, \quad v = x + 1$$

Then:

$$u' = 2x, \quad v' = 1$$

Applying quotient rule:

$$\begin{aligned}
\frac{dy}{dx} &= \frac{(x+1)(2x) - (x^2-1)}{(x+1)^2} \\
&= \frac{2x^2 + 2x - x^2 + 1}{(x+1)^2} \\
&= \frac{x^2 + 2x + 1}{(x+1)^2} \\
&= \frac{(x+1)^2}{(x+1)^2} \\
&= 1
\end{aligned}$$

$$\boxed{\frac{dy}{dx} = 1}$$

9.17 Practice Questions

Basic Problems

1. Differentiate $x^2 + 5x$
2. Differentiate $x^3 - 2x + 1$
3. Differentiate $(x+1)(x^2+2)$
4. Differentiate $\frac{x^2+1}{x}$
5. Differentiate $(x^2+1)^3$
6. Differentiate $\sin x + x^2$
7. Differentiate $e^x + x^3$
8. Find derivative using first principle:

$$f(x) = x^2 + 2x$$

9.18 Increasing and Decreasing Functions

The derivative helps us determine whether a function is increasing or decreasing.

9.18.1 Definitions

- A function is called an **increasing function** in an interval if:

$$f'(x) > 0$$

  for all values of x in that interval.

- A function is called a **decreasing function** in an interval if:

$$f'(x) < 0$$

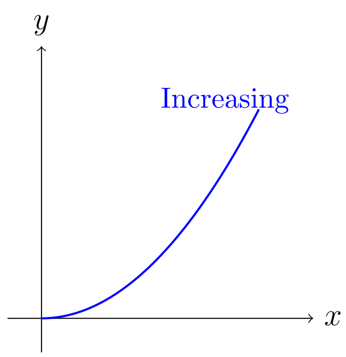
  for all values of x in that interval.

- If:

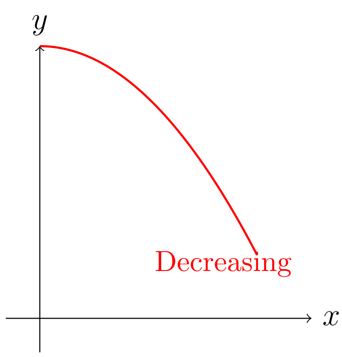
$$f'(x) = 0$$

  then the function may have a maximum point, minimum point, or stationary point.

9.18.2 Geometric Meaning



Increasing Function



Decreasing Function

9.18.3 Test for Increasing and Decreasing Functions

Derivative Test

$$f'(x) > 0 \Rightarrow \text{Increasing Function}$$

$$f'(x) < 0 \Rightarrow \text{Decreasing Function}$$

$$f'(x) = 0 \Rightarrow \text{Stationary Point}$$

Example 1

Determine whether the function

$$f(x) = x^2$$

is increasing or decreasing.

Solution

Differentiate:

$$f'(x) = 2x$$

Now:

- When:

  $$x < 0$$

  then:

  $$f'(x) < 0$$

  Hence, the function is decreasing.

- When:

  $$x > 0$$

  then:

  $$f'(x) > 0$$

  Hence, the function is increasing.

Therefore:

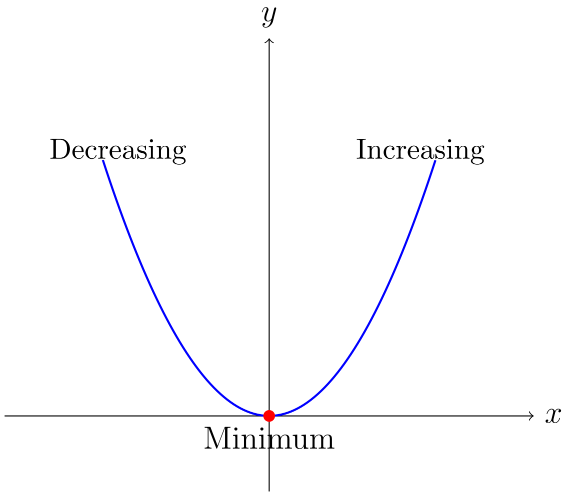
$$f(x) = x^2$$

is decreasing on:

$$(-\infty, 0)$$

and increasing on:

$$(0, \infty)$$



Example 2

Determine the intervals where the function

$$f(x) = x^3 - 3x$$

is increasing or decreasing.
Solution
Differentiate:

$$f'(x) = 3x^2 - 3$$

$$= 3(x^2 - 1)$$

$$= 3(x - 1)(x + 1)$$

Critical points:

$$x = -1, \quad x = 1$$

Now check sign of $f'(x)$:

Interval	Sign of $f'(x)$
$x < -1$	Positive
$-1 < x < 1$	Negative
$x > 1$	Positive

Therefore:

- Increasing on:
$$(-\infty, -1) \cup (1, \infty)$$
- Decreasing on:
$$(-1, 1)$$

Practice Questions

Determine the intervals where the following functions are increasing or decreasing:

1. $$f(x) = x^3$$

2. $$f(x) = x^2 - 4x$$

3. $$f(x) = x^3 - 6x^2$$

4. $$f(x) = \sin x$$

5. $$f(x) = x^4 - 2x^2$$

Important Note

The sign of the derivative determines the behavior of a function.

- Positive derivative $\Rightarrow$ function rises upward.
- Negative derivative $\Rightarrow$ function falls downward.
- Zero derivative $\Rightarrow$ turning point or stationary point.

9.19 Second Derivative Test

The second derivative test is used to determine whether a stationary point is a:

- Maximum point
- Minimum point
- Point of inflection

9.19.1 Theory

Suppose:

$$y = f(x)$$

First find:

$$f'(x)$$

Then solve:

$$f'(x) = 0$$

to obtain stationary points.
Now compute the second derivative:

$$f''(x) = \frac{d^2y}{dx^2}$$

9.19.2 Second Derivative Test

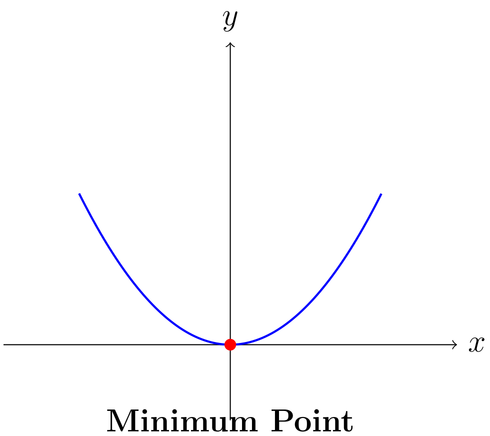
Second Derivative Test

- If:
$$f''(x) > 0$$
then the function has a **minimum point**.

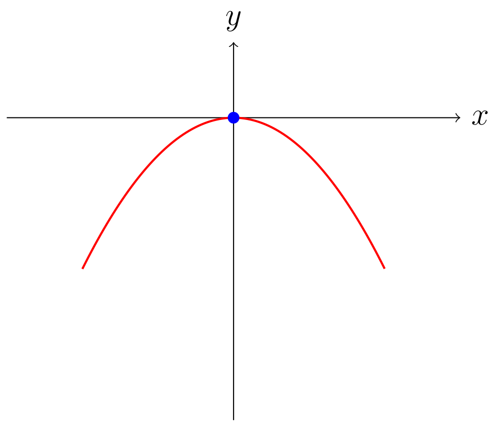
- If:
$$f''(x) < 0$$
then the function has a **maximum point**.

- If:
$$f''(x) = 0$$
then the test fails and further investigation is needed.

9.19.3 Geometric Interpretation



Minimum Point



Maximum Point

Example 1: Minimum Point

Determine whether the stationary point of

$$f(x) = x^2 + 4x + 1$$

is maximum or minimum.
Solution
First derivative:

$$f'(x) = 2x + 4$$

Set equal to zero:

$$2x + 4 = 0$$

$$x = -2$$

Now find second derivative:

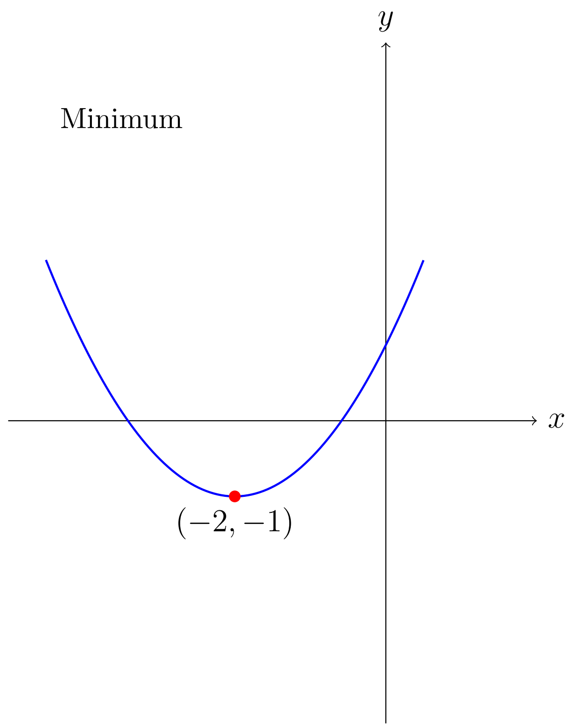
$$f''(x) = 2$$

Since:

$$f''(-2) = 2 > 0$$

Therefore, the function has a:

$$\boxed{\text{Minimum Point at } x = -2}$$



Example 2: Maximum Point

Determine whether the stationary point of

$$f(x) = 4 - x^2$$

is maximum or minimum.
Solution
First derivative:

$$f'(x) = -2x$$

Set equal to zero:

$$-2x = 0$$

$$x = 0$$

Second derivative:

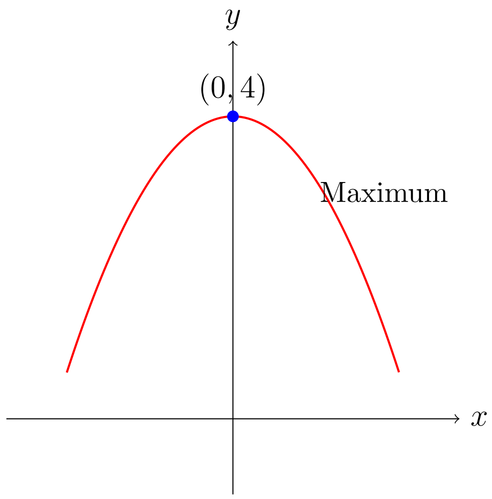
$$f''(x) = -2$$

Since:

$$f''(0) = -2 < 0$$

Therefore, the function has a:

$$\boxed{\text{Maximum Point at } x = 0}$$



Example 3: Point of Inflection

Consider:

$$f(x) = x^3$$

First derivative:

$$f'(x) = 3x^2$$

Second derivative:

$$f''(x) = 6x$$

At:

$$x = 0$$

$$f''(0) = 0$$

Hence, second derivative test fails.
This point is called a:

$$\boxed{\text{Point of Inflection}}$$

Important Notes

- First derivative gives stationary points.
- Second derivative determines nature of stationary point.
- Positive second derivative $\Rightarrow$ Minimum point.
- Negative second derivative $\Rightarrow$ Maximum point.
- Zero second derivative $\Rightarrow$ Test fails.

Practice Questions

Use second derivative test to determine maxima or minima for:

1.
$$f(x) = x^2 - 6x + 5$$

2.
$$f(x) = 9 - x^2$$

3.
$$f(x) = x^3 - 3x$$

4.
$$f(x) = x^4 - 4x^2$$

5.
$$f(x) = 2x^2 + 8x$$

9.20 Chapter Summary

Quick Summary

- Differentiation measures rate of change.
- First principle:
$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$
- Product rule:
$$(uv)' = uv' + vu'$$
- Quotient rule:
$$\left(\frac{u}{v}\right)' = \frac{vu' - uv'}{v^2}$$
- Chain rule:
$$\frac{dy}{dx} = f'(g(x))g'(x)$$